

*Innovations for Advanced Imaging*

*UIS*  
UNIVERSAL  
INFINITY SYSTEM



# Optics

- Expanded range of Olympus UIS optics
- Simple V-shaped optical path maximizes light throughput
- Improved fluorescence illuminator for brighter images
- Newly designed high S/N fluorescence filter cubes
- Improved condensers with high N.A. and longer working distances
- DIC prisms optimized for different specimen types
- Relief Contrast condenser for plastic sample vessels

# System Flexibility

- Unique two-tiered multi-port design offers input/output system flexibility
- Maximum free space out of the left side port
- Small footprint and unique back ports for maximum bench space
- Modular motorized accessories for automation and remote operation
- Sextuple nosepiece and fluorescence filter turret

# A new standard in inverted microscopy

**Stable compact frame, two-tiered multi-port design and outstanding UIS optical performance for imaging, measurement and micromanipulation**

The IX71 microscope system begins with a highly rigid, compact and stable frame. Thermal and structural rigidity have been increased by utilizing an external power supply and computer aided design techniques. Unique two-tiered multi-port design and a small footprint combine for maximum system flexibility while conserving critical bench space. Improved fluorescence illuminators and new high S/N ratio cubes combined with a V-shaped optical path result in very bright, low noise fluorescence images. Low stage and focus controls, high visibility setting indicators, front panel operation and a tilting binocular tube combine to provide exceptional ease of use and comfort. IX71: setting new standards in research microscopy.

*\*Note:*

*UIS optics: The original Olympus infinity-corrected optical system.*

*This system expresses the superb Olympus optical technology, provides high resolution, high contrast images and the flexibility to meet current and future applications.*

# Applications

- Time-lapse:  
Rigid and stable frame maintains steady specimen focus
- Fluorescence:  
Chromatically corrected illumination from UV to IR
- Microinjection:  
Relief Contrast for injection in plastic vessels
- Multi Dimensional Imaging:  
Multiple simultaneous image ports
- TIRFM:  
Special illuminator for evanescent wave illumination



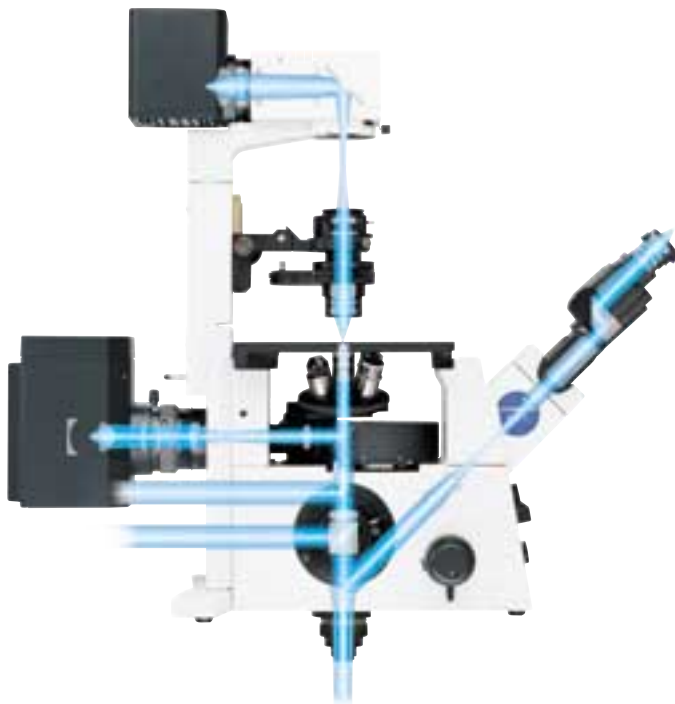
# Unique two-tiered multi-port design offers input/output system flexibility and application support

The IX frame begins with 6 input/output ports for a wide variety of light sources or cameras. The unique Olympus design allows up to four ports to have simultaneous access to the primary image, without relay lenses, providing cameras with the highest quality image, free of aberrations. Two-tiered multi-port design maximizes flexibility while maintaining a 299mm low stage height. The compact and rigid frame requires a minimum amount of bench space, easily accommodates external equipment and remains stable during extended observations. Over 100mm of free space is available at the left side port between the frame and the primary image, easily allowing for adaptation of instrumentation for individual research requirements.

*Multi-port system*

*Clear primary image*

*System versatility*



**299mm low stage height**

UPPER BACK PORT

LOWER BACK PORT

LEFT SIDE PORT

**102mm of free space**

**394mm in depth**

BOTTOM PORT



**A B C D**

**Primary images are obtainable from all 4 ports simultaneously**



### LEFT SIDE PORT (Primary Image)

**A B**

For film or CCD camera systems. Double port tube allows the attachment of two cameras (both primary images). Also accepts Olympus confocal scan units.

U-DPCAD double port tube with C-mounts on left side port



### RIGHT SIDE PORT (Primary Image)

**C**

Used for connecting an additional light source or CCD camera.

IX2-RSPC right side port attachment with C-mount



### BOTTOM PORT (Primary Image)

**D**

Well suited for extremely sensitive low light level imaging.

IX-TVR bottom port unit



### UPPER BACK PORT (Primary Image)

Used for connecting an additional light source or CCD camera.

### LOWER BACK PORT (Primary Image)

For the attachment of large bodied items such as cooled CCD cameras or confocal scan units.



### BINOCULAR PORT

CCD camera mount via side port intermediate tube.

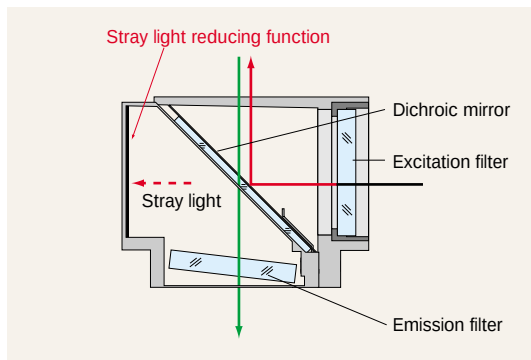


GX-SPU side port intermediate tube

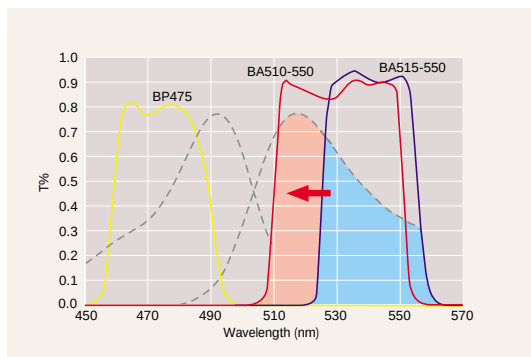
Side port intermediate tube for CCD camera.

# New fluorescence accessories and V-shaped optical path for improved sensitivity in critical applications

The very efficient Olympus mercury and xenon lamphouses are combined with fluorescence illuminators designed for utmost flexibility and ease of use. An L-shaped illuminator provides easy access to burner centration and removable aperture and field stops. An improved straight illuminator has greater throughput and also accepts a new dual port adapter for simple switching between two light sources. Fluorescence filter cube performance has been improved with high efficiency coatings optimized for specific fluorochromes such as GFP, YFP and CFP, with minimal signal crossover. Stray light reducing materials are incorporated into all fluorescence filter cubes.



When the fluorescence excitation light is reflected by the dichroic mirror, some light may pass through the mirror and reflect as stray light in the cube assembly. This stray light may reflect out of the cube and into the imaging path, resulting in decreased contrast due to autofluorescence. Low reflection materials are incorporated into the design of all new Olympus fluorescence filter cubes. Such a design absorbs over 99% of all stray light, resulting in high contrast fluorescence images.



The new IBA cube is optimized to maximize both GFP excitation and detection of GFP emission through the use of a new dichroic mirror with minimal crossover.

**V-shaped optical path for maximum light throughput**



*High Contrast  
fluorescence*

*Wide range of  
accessories*

*Utmost flexibility*

**Aspherical apochromat  
collector lens**

**Exchangeable field  
stop/aperture stop**

**6 position cube turret**

**Newly designed high S/N  
fluorescence filter cubes**

**Lamphouses with aspherical collector lenses/  
U-LH100HGAPO, U-LH75XEAP0**

An aspherical collector lens design in the mercury and xenon arc lamphouses improves light collection ability and stretches the apochromatic performance of the illumination optics to include wavelengths from UV to IR.



**L-shaped reflected light illuminator/IX2-RFAL**

Provides easy access to burner centration and removable aperture and field stops. The L-shaped design maintains access to both back frame ports.



**Rectangular field stop/IX2-RFSS  
Pinhole field stop/IX2-RFSPOT**

Flexible field stop options IX2-RFSS rectangular field stop module and IX2-RFSPOT pinhole field stop module can be mounted in the L-shaped illuminator for better signal to noise ratio fluorescence imaging or photobleaching experiments.



**Reflected light illuminator/IX2-RFA**

Straight type illuminator designed for maximum throughput is 20% brighter than the previous model. Well suited for applications requiring high intensity excitation or multiple excitation filters.



**Double lamp housing adapter/U-DULHA**

Allows simultaneous attachment of two light sources such as halogen and mercury. Selection mirror is replaceable for custom applications.

(Available autumn 2002)



**Long working distance objective/LUCPLFL40x**

Accommodating vessel thicknesses from 0-2mm with a correction collar mechanism that maintains objective focus during adjustment. Correcting for different vessels is fast and simple. Lens design is also optimized for 340nm transmission for calcium ratio imaging.



# Improved thermal and structural rigidity combine to provide a stable platform for time-lapse observations

From the computer engineered compact frame to the external power supply, the IX71 was designed for structural and thermal stability. The nosepiece stage provides an extremely stable platform for time-lapse experiments by maintaining objective focus. The potentially negative effects of heat and electrical noise have been removed by employing a radiantly cooled, external power supply for transmitted light. Microscope controls, such as light path selectors and mirror unit turrets are designed for shock-free operation. Microscope frame and condenser system accommodates incubation and CO<sub>2</sub> chambers for observations requiring a constant environment.



## Nosepiece stage/IX2-NPS

Maintains objective focus and essentially eliminates focus-drift during time-lapse and prolonged observations.

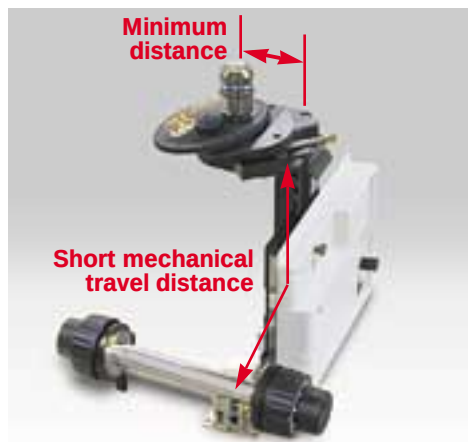


**Maximum  
rigidity & stability**

**Low center of  
gravity design**

**Shock-free  
operation**

**Maximum rigidity and  
minimum thermal  
expansion**



Short mechanical circuit structure and high rigidity nosepiece design avoids pitching and yawing when the nosepiece is manipulated during operations such as DIC prism insertion or adjustment of an objective's correction collar.

**Low stage, low center  
of gravity design**

**External power  
supply to prevent  
thermal expansion**

## Frontal control allows auxiliary equipment to be placed near the microscope

A key problem when conducting experiments on living cells is the placement of auxiliary equipment. The IX71 is specifically designed to make the fullest use of desktop space by employing such features as a back port and a detachable wing. Even while operating the focusing knobs, the operator can easily perform functions such as changing the light path, light intensity adjustment, shutter ON/OFF and power ON/OFF by means of buttons or levers conveniently located on the front panel. Operability is further improved by a tilting observation tube which lets the operator make observations while standing or sitting.

**Tilting  
binocular tube**

**Frontal  
control**



Optional tilting binocular tube enables comfortable observations when sitting or standing.

Front mounted microscope controls for improved ergonomic comfort also permit access to additional equipment at the sides of the microscope.



# Excellent condenser design and versatility accommodates a wide range of sample vessels and applications

For transmitted light applications, the IX2 condenser system offers both high resolution and high contrast images for a wide range of specimens and sample vessels.

Olympus introduces the Relief Contrast system, providing DIC-type images with plastic vessels.

Relief Contrast, combined with the stability of the IX frame, results in an excellent injection/manipulation workstation.

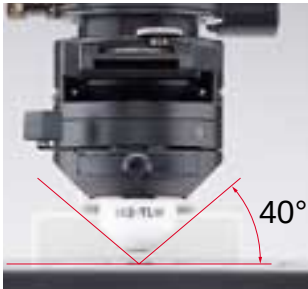
For electrophysiology, the water immersion condenser provides a high numerical aperture for image quality while maintaining a long working distance and an optimal 40° access angle. For incubation chamber and T-flask observations a longer working distance, high numerical aperture condenser provides excellent phase and DIC images.

## Electrophysiology

### IX2-DICD

| N.A. | W.D.  | PH | DIC |
|------|-------|----|-----|
| 0.9  | 3.7mm |    |     |

The slim design of the DICD condenser employs a single DIC slider combined with a tapered front lens for stability and maximum access angles. The result is excellent operability for injection and patch clamping experiments. Dry and oil top lenses are also available for versatility with different observation techniques and sample preparations. Phase contrast condenser inserts are also available.



## IVF/Injection

### IX2-MLWCD

| N.A. | W.D. | RC | PH | DIC |
|------|------|----|----|-----|
| 0.5  | 45mm |    |    |     |

The Relief Contrast condenser is designed to produce contrast and shading effects, similar to DIC, yet within the confines of plastic sample vessels. The unique ability to adjust the illumination angle and shading effect of each objective provides consistent illumination shadowing across all magnifications.

To facilitate injections, the contrast effect on cellular membranes has been optimized.



## Incubation Chambers/T-Flasks

### IX2-LWUCD

| N.A. | W.D. | PH | DIC |
|------|------|----|-----|
| 0.55 | 27mm |    |     |

Combining a long working distance (27mm) and a high numerical aperture (N.A. 0.55), the LWUCD condenser accommodates most incubation chambers and T-Flasks.

The 5-position turret provides versatility with DIC or phase inserts. DIC components are specially designed to obtain high-contrast, high-resolution images with 20X and 40X objectives.



## Ultra-long Working Distance

### IX-ULWCD

| N.A. | W.D. | PH |
|------|------|----|
| 0.3  | 73mm |    |

This condenser accommodates phase contrast and brightfield microscopy with 4X, 10X, 20X and 40X objectives. Excellent image contrast is assured for any sample — thin to thick cells.

An extended working.



# Modular system easily accommodates a wide range of research applications

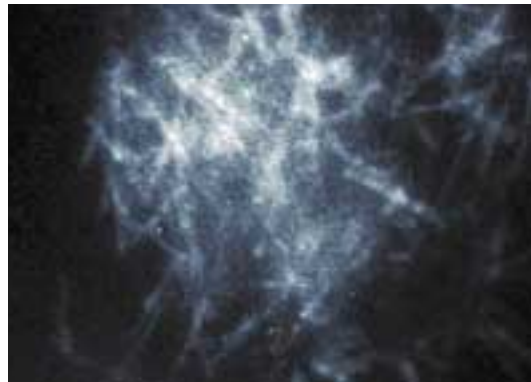
## *Micromanipulation*

Low stage design combined with a lowered center of gravity offers superior stability for micromanipulation applications. A variety of screw holes are available to securely mount manipulators onto the microscope, allowing optimal choice of angles and positioning. Manipulator joysticks can be placed in the most comfortable, accessible position due to the compact frame design.



## *TIRFM*

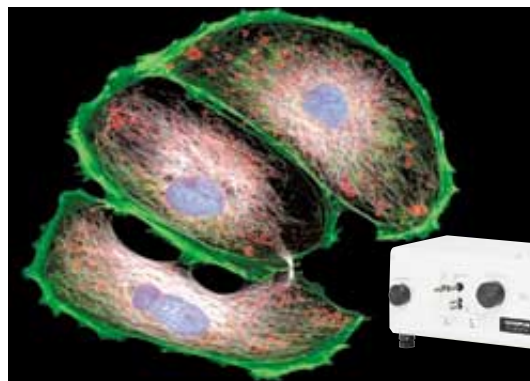
Total Internal Reflection Fluorescence illuminator provides easy setup and alignment of excitation laser light. Mirror unit allows switching between fluorescence and TIRF illumination and can be customized. High numerical aperture APO100XOHR (N.A. 1.65) and PLAPO60XOTIRFM (N.A. 1.45) objectives offer the brightest signal and greatest adjustment range for TIRFM applications.



## *Laser Scanning Confocal*

Frame easily accepts Olympus Fluoview Confocal microscope system for high resolution, low noise observations. Back or left side port accepts confocal scan units. Stable, rigid frame is ideal for laser input/output, z-stacks, and line scanning. Optional motorized accessories can be added to create a powerful confocal workstation.

\* Special microscope body available for Fluoview.



# Optional units

## Motorized fluorescence cube turret/IX2-RFACA

Accepts up to 6 fluorescence filter cubes.



## Motorized observation/emission side filter wheel/ U-FWO

### Motorized reflected/excitation side filter wheel/U-FWR

Up to 6 filters (32mm or 25mm diameter) can be attached.



## Motorized long working distance universal condenser/IX2-LWUCDA

6 turret positions allow brightfield, phase contrast and DIC observations (N.A. 0.55, W.D. 27mm).



## Motorized bottom port unit with C-mount/ IX2-TVRAC

Entirely aberration-free primary images from UIS objectives are directed to a c-mount CCD camera.



## Gliding stage/IX2-GS

Designed for applications such as *C. elegans* observation. Stage rotates a full 360° with 20mm X-Y travel. The flat design allows horizontal mount of micromanipulators.



## Glass stage insert plate/ IX2-GCP

Includes glass plate and colored band system for easy objective magnification confirmation.

\* Not available in some areas



## Plain stage (IX2-SP)+ Mechanical stage (IX-MVR)

Plain stage plus an optional mechanical stage provide X-Y movement for a variety of sample vessels. Coaxial control knobs are positioned low for user comfort.

### IX-MVR accessories:

- Terasaki plate holder/IX-HOT  
Accepts 60 and 72-well Terasaki plates
- Slide glass holder/IX-HOS  
Designed for 1in x 3in slides or hemocytometer
- Petri dish holder/IX-HOP  
Holds 65mm, 54mm and 35mm diameter dishes
- Well positioner T/IX-CLMT  
Provides click stops for well-center of 60 and 72-well Terasaki plates
- Well positioner 96/IX-CLM96  
Provides click stops for well-center of 96-well plate
- Millimeter scale/IX-PPM  
Vernier scale for stage in millimeters
- Scale for nunc plate/IX-PP24NUN  
Scales for NUNC style 24-well plate



## Heat plate

Compact automatic temperature control unit that replaces stage insert. Available configurations: Standard (accurate to  $\pm 3^{\circ}\text{C}$ ); High-grade (accurate to  $\pm 0.1^{\circ}\text{C}$ ); and Reduced noise (accurate to  $\pm 0.1^{\circ}\text{C}$ ).

\* Tokai Hit Company products



## CO2 incubator

Enables highly accurate control of temperature, moisture and carbon dioxide. Ideal for prolonged time-lapse observations such as cell division.

\* Tokai Hit Company products



# UIS series objectives



## LCPLFL series

These Semi-Apochromat objectives are dedicated for tissue culture observations through bottles and dishes, offering excellent contrast and resolution in brightfield, DIC and fluorescence observations.



## LCPLFL-PH series

These objectives are exclusively designed for culture specimens. Thanks to the correction cap method, an excellent phase contrast image is assured regardless of the thickness and material of the vessel.



## UPLFL series

These affordable Semi-Apochromat universal objectives deliver superb resolution, contrast and flatness for any microscopy technique.



## UPLFL-PH series

The newly designed phase annuli reduce flare and halo to a minimum and ensure high resolution and contrast for unstained specimens, i.e. living cells and microorganisms.



## UPLAPO series

Top-performance universal Plan Apochromat objectives, featuring an unsurpassed high N.A., deliver the best resolution, contrast and field flatness for any microscopy technique.



## PLAPO series

These Plan Apochromat objectives keep chromatic aberration down to an absolute minimum and deliver the best resolution and contrast for brightfield, DIC and fluorescence observations.



## UAPO/340 series

These objectives feature high transmission of 340nm wavelength light, ensuring maximum performance in fluorescence microscopy through UV excitation including  $\text{Ca}^{2+}$  measurement.



## Relief Contrast objectives

These objectives are designed for observation of living cells including oocyte. Plastic vessels applicable for Relief Contrast observation.

| Item                     | N.A.      | W.D.<br>(mm) | Cover glass<br>thickness (mm) |
|--------------------------|-----------|--------------|-------------------------------|
| CPLFL10XPH               | 0.30      | 9.5          | 1                             |
| LCPLFL20X                | 0.40      | 6.9          | 0-2.5                         |
| LCPLFL20XPH              | 0.40      | 6.9          | 0-2.5                         |
| LCPLFL40X                | 0.60      | 2.6          | 0-2.5                         |
| LCPLFL40XPH              | 0.60      | 2.6          | 0-2.5                         |
| LCPLFL60X                | 0.70      | 1.7          | 0-2.5                         |
| LCPLFL60XPH              | 0.70      | 1.7          | 0-2.5                         |
| LUCPLFL40X               | 0.60      | 2.7-4.1      | 0-2                           |
| LUCPLFL40XPH             | 0.60      | 3.0-4.3      | 0-2                           |
| CPL10XPH                 | 0.25      | 9.8          | 1                             |
| LCACH20XPH               | 0.40      | 3.2          | 1                             |
| LCACH40X2PH              | 0.55      | 2.3          | 1                             |
| UPLFL4X                  | 0.13      | 17.0         | —                             |
| UPLFL4XPH                | 0.13      | 17.0         | —                             |
| UPLFL10X                 | 0.25      | 10.0         | —                             |
| UPLFL10XPH               | 0.30      | 10.0         | —                             |
| UPLFL20X                 | 0.50      | 1.6          | 0.17                          |
| UPLFL20XPH               | 0.50      | 1.6          | 0.17                          |
| UPLFL40X                 | 0.75      | 0.51         | 0.17                          |
| UPLFL40XO                | 1.30      | 0.1          | 0.17                          |
| UPLFL40XPH               | 0.75      | 0.51         | 0.17                          |
| UPLFL60XOI3              | 0.65-1.25 | 0.1          | 0.17                          |
| UPLFL60XOI3PH            | 0.65-1.25 | 0.1          | 0.17                          |
| UPLFL100XOI3             | 1.30      | 0.1          | 0.17                          |
| UPLFL100XOI3PH           | 0.60-1.3  | 0.1          | 0.17                          |
| UPLFL100XOI3PH           | 1.3       | 0.1          | 0.17                          |
| UPLAPO10X                | 0.40      | 13.0         | 0.17                          |
| UPLAPO10XOI3             | 0.40      | 0.17         | —                             |
| UPLAPO10XW3              | 0.40      | 0.44         | 0.17                          |
| UPLAPO20X                | 0.70      | 0.65         | 0.17                          |
| UPLAPO20XOI3             | 0.80      | 0.19         | —                             |
| UPLAPO40X                | 0.85      | 0.20         | 0.11-0.23                     |
| UPLAPO40XOI3             | 0.5-1.0   | 0.12         | —                             |
| UPLAPO60XW3              | 1.20      | 0.25         | 0.15-0.21                     |
| UPLAPO60XW3/IR           | 1.20      | 0.28         | 0.13-0.21                     |
| UPLAPO60XW/PSF           | 1.20      | 0.25         | 0.15-0.21                     |
| UPLAPO100XOI3            | 0.5-1.35  | 0.1          | 0.17                          |
| PLAPO60XOI3              | 1.40      | 0.15         | 0.17                          |
| PLAPO60XOTIRFM           | 1.45      | 0.15         | 0.17                          |
| APO100XOHR <sup>*1</sup> | 1.65      | 0.1          | 0.15                          |
| UAPO20X3/340             | 0.75      | 0.55         | 0.17                          |
| UAPO20XW3/340            | 0.70      | 0.35         | 0.17                          |
| UAPO40X3/340             | 0.90      | 0.2          | 0.11-0.23                     |
| UAPO40XOI3/340           | 0.65-1.35 | 0.1          | 0.17                          |
| UAPO40XW3/340            | 1.15      | 0.25         | 0.13-0.25                     |
| CPLFL10XRC               | 0.30      | 9.1          | 1.5 <sup>*2</sup>             |
| CPL10XRC                 | 0.25      | 9.6          | 1.5 <sup>*2</sup>             |
| LCPLFL20XRC              | 0.40      | 6.9          | 1.5 <sup>*2</sup>             |
| LUCPLFL40XRC             | 0.60      | 3.0-4.3      | 0-2                           |
| LCACH20XRC               | 0.40      | 2.8          | 1.5 <sup>*2</sup>             |
| LCACH40XRC               | 0.55      | 2            | 1.5 <sup>*2</sup>             |

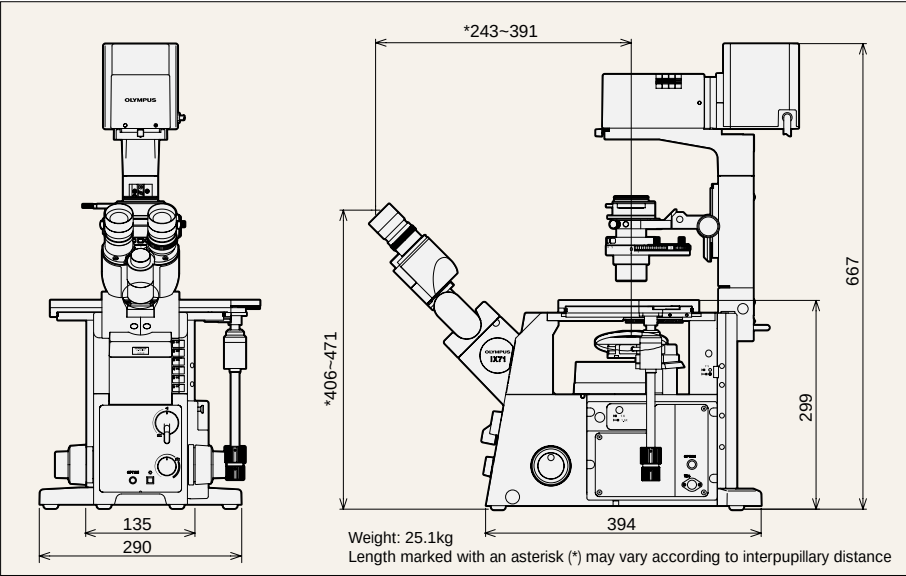
<sup>\*1</sup> Special oil and special cover glass required

<sup>\*2</sup> Optically designed to be the best use with TOKAI-HIT's glass heat plate with 0.5mm thickness

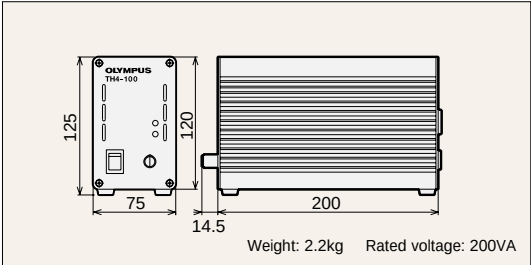
IX71 specifications

|                                   |                                            |                                                                                       |                                                                                                                                                                                                                                                                                                                |
|-----------------------------------|--------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Microscope body                   | Revolving nosepiece                        |                                                                                       | Sextuple, simple waterproof mechanism incorporated                                                                                                                                                                                                                                                             |
|                                   | Focus                                      |                                                                                       | 9mm stroke (from stage surface, 7mm upward and 2mm downward), coaxial coarse and fine focusing knobs (minimum fine focus graduation: 1µm, full rotation of fine focusing knobs: 100µm), upper limit stopper, torque adjustment for coarse focusing                                                             |
|                                   | Primary image port                         |                                                                                       | Lower port (standard left side port: S1F 100% or S8F 80%, or optional lower Back port selectable, 2-step light path selection), Upper port when built-in magnification changer 1x/1.6x is replaced (optional right side port or upper back port selectable, 2-step light path selection), Bottom port (option) |
|                                   | Frontal operation                          |                                                                                       | Light path selector, Transmitted light intensity control and light ON/OFF switch, TTL Pulse control switch                                                                                                                                                                                                     |
| Transmitted light illuminator     | 100W transmitted light illumination pillar | IX2-ILL100                                                                            | Pillar tilt mechanism (30° inclination angle, with shock decreasing mechanism), Condenser holder (with 50mm stroke, swing-in/out mechanism), Field iris diaphragm adjustable, 4 filter holders (ø45mm, t=6mm or less)                                                                                          |
|                                   | External power supply unit                 | TH4-100/200                                                                           | Two versions available (100V and 200V), Optional TH4-HS hand switch can be used, 2.2kg weight                                                                                                                                                                                                                  |
| Observation tube                  | Tilting binocular tube                     | U-TB/90                                                                               | 35-85° continuous angle adjustable (eyepoint height range: 406mm-471mm), interpupillary distance adjustable between 50-76mm, diopter adjustment function, erect image, F.N. 22                                                                                                                                 |
|                                   | Binocular tube                             | U-BI90CT                                                                              | Built-in focusing telescope, interpupillary distance adjustable 50-76mm, diopter adjustment function, F.N. 22                                                                                                                                                                                                  |
|                                   |                                            | U-BI90                                                                                | Interpupillary distance adjustable 50-76mm, diopter adjustment function, F.N. 22                                                                                                                                                                                                                               |
|                                   | Trinocular tube                            | TR30H+IX-ATU                                                                          | 3 step optical path selectable (observation : straight port = 100:0, 20:80, 0:100), interpupillary distance adjustable 50-76mm, diopter adjustment function, F.N. 22                                                                                                                                           |
| Stage                             | Cross stage with flexible right handle     | IX2-SFR                                                                               | 50mm(X) x 50mm(Y) stroke, stage insert plate exchangeable (ø110mm)                                                                                                                                                                                                                                             |
|                                   | Cross stage with short left handle         | IX-SVL-2                                                                              | 50mm(X) x 43mm(Y) stroke, stage insert plate exchangeable (ø110mm)                                                                                                                                                                                                                                             |
|                                   | Plain stage                                | IX2-SP                                                                                | 232mm(X) x 240mm(Y) stage size, stage insert plate exchangeable (ø110mm)                                                                                                                                                                                                                                       |
|                                   |                                            | IX-MVR                                                                                | Mechanical stage to be used with IX2-SP, 130mm(X) x 85mm(Y) stroke                                                                                                                                                                                                                                             |
|                                   | Narrow plain stage                         | IX2-KSP                                                                               | 160mm(X) x 240mm(Y) stage size, stage insert plate exchangeable (ø110mm)                                                                                                                                                                                                                                       |
|                                   |                                            | CK40-MVR                                                                              | Mechanical stage to be used with IX2-KSP, 120mm(X) x 78mm(Y) stroke                                                                                                                                                                                                                                            |
|                                   | Gliding stage                              | IX2-GS                                                                                | Upper circular stage 360° rotatable, 20mm(X/Y) travel                                                                                                                                                                                                                                                          |
| Condenser                         | Long working distance universal            | IX2-LWUCD                                                                             | 5 positions for optical devices (3 positions for ø30mm and 2 position for ø38mm), aperture iris diaphragm adjustable, N.A. 0.55 / W.D. 27mm                                                                                                                                                                    |
|                                   | Long working distance Relief Contrast      | IX2-MLWCD                                                                             | 4 positions for optical devices (for ø50mm, Relief Contrast optical devices rotatable), aperture iris diaphragm adjustable, N.A. 0.5 / W.D. 45mm                                                                                                                                                               |
|                                   | Ultra long working distance                | IX-ULWCD                                                                              | 4 positions for optical devices (for ø29mm), aperture iris diaphragm adjustable, N.A. 0.3 / W.D. 73mm                                                                                                                                                                                                          |
|                                   | Water immersion DIC                        | IX2-DICD + IX2-TLW                                                                    | Single position for optical device (include two optical device holders), 40° injection pipette or electrode insertion angle, aperture iris diaphragm adjustable, N.A. 0.9 / W.D. 3.7mm                                                                                                                         |
| Eyepiece                          |                                            | WH10x                                                                                 | High eyepoint, F.N. 22                                                                                                                                                                                                                                                                                         |
|                                   |                                            | WH10x-H                                                                               | High eyepoint, diopter adjustment function, F.N. 22                                                                                                                                                                                                                                                            |
| Reflected light fluorescence unit | Fluorescence illuminator                   | IX2-RFAL                                                                              | L-shaped design with exchangeable F.S. and A.S. modules, two filter holder sliders (2 positions, ø32mm, t=6mm or less)                                                                                                                                                                                         |
|                                   |                                            | IX2-RFA                                                                               | Straight design with field iris diaphragm, filter holder slider (2 positions, ø32mm, t=6mm or less)                                                                                                                                                                                                            |
|                                   | Fluorescence cube turret                   | IX2-RFAC                                                                              | 6 positions in a rotating turret, built-in shutter                                                                                                                                                                                                                                                             |
|                                   | Light source                               | 100W HBO lamp housing and HBO transformer, or 75WXBO lamp housing and XBO transformer |                                                                                                                                                                                                                                                                                                                |

IX71 dimensions



TH4 dimensions



TH4-HS dimensions

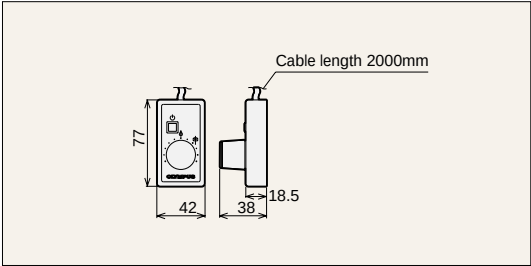


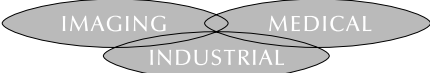
Photo courtesy of: Yuji Abe M.D. Ph.D., The 1st Department of Obstetrics & Gynecology School of Medicine Toho University (P.10 above IVF)

Specifications are subject to change without any obligation on the part of the manufacturer.



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